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SOLAR ESS TECHNICAL REPORT · 1 NOVEMBER 2026 · REV01

Electrical Air Clearance & Heat Dissipation

What the topic covers, why it matters, how to get it right on solar, battery and switchboard installations, and what the evidence from real incidents shows.

A. Air clearance

`space for VOLTAGE`

Air gap and surface path between live parts, and to earth — stops flashover and tracking.

B. Working space

`space for PEOPLE`

Clear zone in front of boards, inverters and batteries — safe work, escape, doors open 90°.

C. Heat clearance

`space for AIR`

Gaps around inverters, batteries, boards and cables — lets heat leave so parts stay in rating.

The word "clearance" is used for three different jobs. This report covers all three because they fail together: a cramped installation is usually short of all of them.

8 mm
min. air clearance for 1000 V equipment, OVC III, ≤2000 m [4]

×2
capacitor life lost for every +10°C [14]

0.38
cable rating factor with 20+ circuits bunched [15]

~25%
of major electrical failures from loose/faulty connections [22]

Key takeaways

- **Air clearance** is set by the voltage spike the equipment must survive, the dirt and moisture around it, and altitude — not by the normal operating voltage alone [1,2].
- **Working space** in Thailand (EIT 2556): at least 0.75 m wide and the width of the equipment, doors opening 90°, 2.0 m headroom [7,8].
- **Heat** is the quiet failure mode: inverters cut output (derate), electronics age faster, cables and terminals run past their ratings [10,14,15].
- **Evidence:** PV fires are rare, but where they happen they start mostly at DC isolators, connectors and other joints — points where heat builds up [17,18,21].

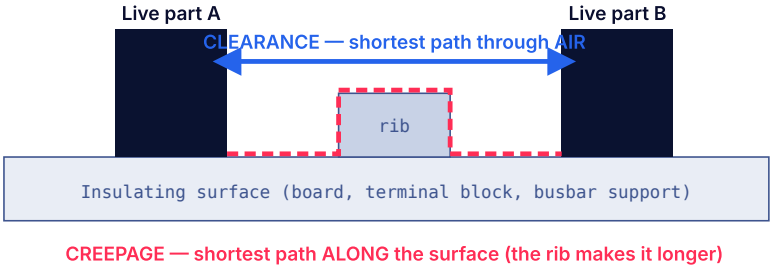
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PART A · SPACE FOR VOLTAGE

Electrical air clearance and creepage

What it is

Clearance is the shortest distance through air between two conductive parts; **creepage** is the shortest distance along the surface of the insulation between them [1,3]. The international rulebook is IEC 60664-1 (insulation coordination for low-voltage equipment), which applies up to 1,000 V AC and 1,500 V DC — the whole range of PV strings, batteries and building wiring [3,4]. Product standards build on it, e.g. IEC 62109-1 for PV inverters and IEC 61439 for switchboards [4].



Clearance protects against flashover through air; creepage protects against “tracking” — a conductive path that slowly forms on a dirty, damp surface.

What decides the distance

Factor	What it means	Effect
Overvoltage category (OVC I–IV)	How large a transient (lightning, switching) the equipment must survive, based on where it sits in the supply [1]	Sets the rated impulse voltage, which drives **clearance**
Pollution degree (PD 1–4)	From clean and dry (PD1) to conductive dust, rain or wetness (PD4) [1]	Drives **creepage**; PD2→PD3 can add 60% or more [2]
Material group (CTI)	How well the plastic resists tracking [2]	Lower-grade plastics need longer creepage
Altitude	Thinner air breaks down more easily; correction above 2,000 m [2]	×1.14 at 3,000 m, ×1.29 at 4,000 m [4]

Worked example [4]: equipment rated 1,000 V, OVC III → rated impulse withstand 8 kV → minimum clearance **8 mm** at or below 2,000 m. Coating a board does not reduce clearance — clearance is always measured through air [2].

Why it matters on site

- The installer cannot change the clearances inside a certified product — but can easily **destroy** them: stray strands of fine-stranded wire, oversized lugs turned towards each other, cables pressed against busbars, or a DC isolator designed for AC [17].
- Thailand's rainy season brings months of high humidity, so outdoor boxes, combiner boxes and rooftop isolators should be treated as dirty-and-damp locations: use enclosures and parts rated for that, keep glands sealed and keep dust out. Surges (lightning, switching) are the real test of clearance, so SPDs and earthing belong to the same picture.

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PART B · SPACE FOR PEOPLE

Working space around equipment

What it is

A clear, dedicated zone in front of equipment that may be examined, adjusted or maintained while energised. Thailand's installation standard (EIT/วสท. 2556) requires it to be at least **0.75 m wide** and not narrower than the equipment, with depth per its Table 1-1, and enough room for doors to open at least **90°** [7]. Where rear access is needed, 0.75 m behind the equipment; headroom at least **2.00 m**; and the top of a switchboard at least **0.90 m** from a combustible ceiling (0.60 m if the ceiling is non-combustible or shielded) [8].



Picture the working space as an empty refrigerator box standing in front of the board: nothing may be stored in it.

International reference: NEC Table 110.26(A)(1) — minimum depth

Voltage to ground	Condition 1 nothing live/earthed opposite	Condition 2 earthed surface opposite (concrete, brick)	Condition 3 live parts on both sides
0–150 V	0.9 m	0.9 m	0.9 m
151–600 V	0.9 m	1.0 m	1.2 m
601–1000 V	0.9 m	1.2 m	1.5 m

Source: NEC 110.26 extract [5]. Width at least 30 in (762 mm) or equipment width; height 6½ ft (2.0 m) [6]. Thai 230/400 V systems fall in the 151–600 V row (230 V to ground).

Batteries: extra separation, not just working space

Australia's battery standard AS/NZS 5139 is the most detailed public reference for home batteries. It forbids batteries within 600 mm of an exit, of a window or vent into a habitable room, or of any appliance, and within 900 mm below those — and not in ceiling spaces or wall cavities [9]. Thailand has no equivalent battery-siting rule yet; Solar ESS uses these distances as good practice.

> **Why it matters:** a fault on a board is most likely during switching or maintenance. Without depth the electrician cannot step back from an arc; without width a door cannot open fully; boxes stored in front of a board block the way to the main switch in an emergency.

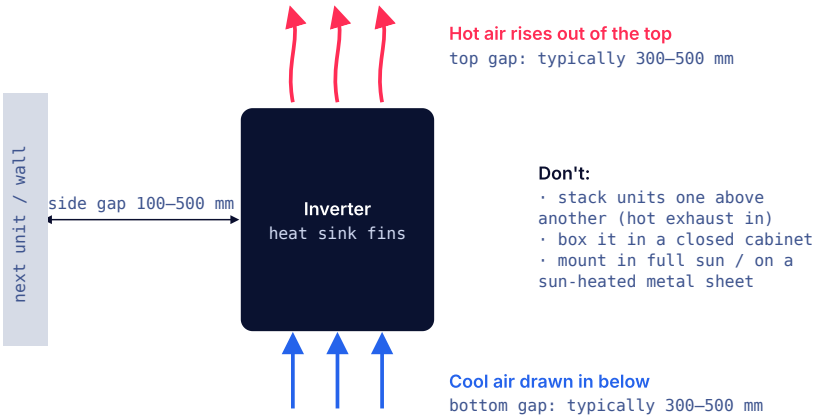
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PART C · SPACE FOR AIR

Heat dissipation clearance

What it is

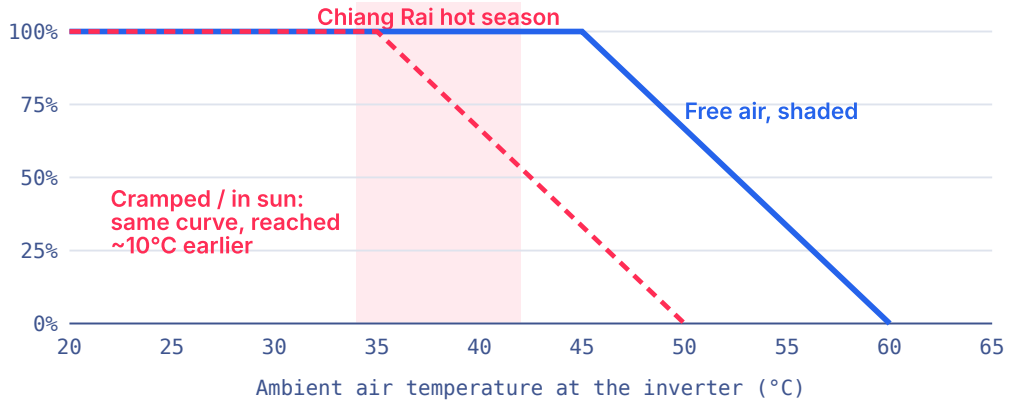
Every electrical part turns some energy into heat: conversion losses inside an inverter, I²R loss in a cable or terminal, charging losses in a battery. Converting DC to AC cannot happen without losses [10]. That heat must leave by conduction, convection and radiation; clearance gaps are what let convection work. Manufacturers state minimum gaps — for example, one inverter range calls for 500 mm on all sides (300 mm on tight sites) [11]; another needs 300 mm above, 500 mm below and 200 mm at the sides [26].



The “chimney”: cool air enters below, passes the fins, and leaves at the top. Block either end and the inverter breathes its own exhaust.

Why it matters — 1. Lost energy (derating)

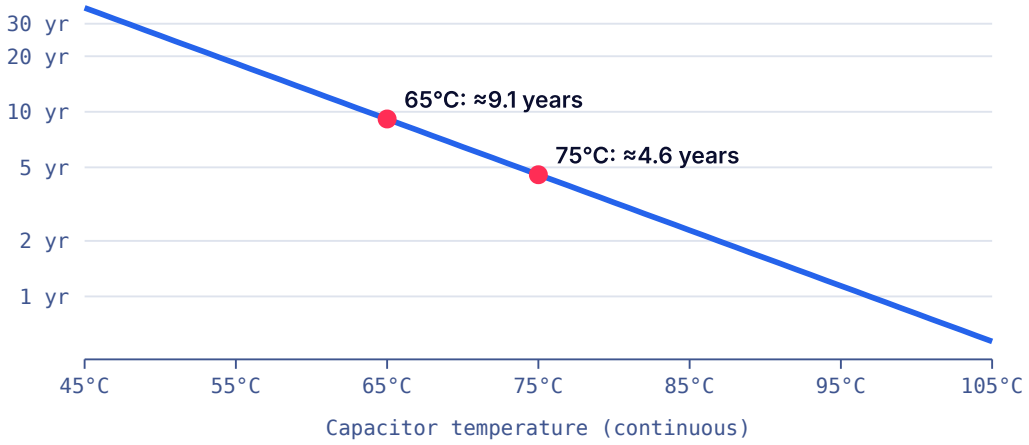
When the semiconductors approach their limit the inverter moves to a lower power point to protect itself — temperature derating [10]. Typical causes are poor installation conditions, direct sun or high ambient temperature [10]. Thresholds vary: one quality 5 kW model starts derating at 50°C ambient; another at 40°C [13]. A sun-heated metal back sheet needs about 30 cm gap, otherwise derating starts earlier than expected [12].



Illustrative curve (shape typical of datasheets; values vary by model). Chiang Rai averages 34°C maximum in April [24]; a cramped or sunny spot adds to that.

Why it matters — 2. Shorter equipment life

Aluminium electrolytic capacitors are usually what limits an inverter's life. Their ageing follows the Arrhenius rule: life halves for every 10°C rise; a 5,000-hour/105°C part lasts about 10,000 h at 95°C and 20,000 h at 85°C [14].



Calculated with $L = 5{,}000\ h \times 2^{(105 - T)/10}$ for continuous operation; real inverters run part-time and use better parts, but the ratio holds: 10°C hotter = half the life.

Why it matters — 3. Cables and boards outside their rating

Cables. Ratings in IEC 60364-5-52 assume 30°C air and a single circuit. Hotter air and bunching both cut the allowed current: 20 or more circuits bunched together keep only **38%** of the single-cable rating [15]. Long DC runs bundled in one conduit on a hot roof are a typical case.

Switchboards. IEC 61439 verifies temperature rise at a 35°C reference ambient; terminals for external cables may rise no more than **70 K** (i.e. 105°C) [16]. A sealed box on a west wall in April is outside that reference, so load, ventilation and spacing of breakers must be checked.

Equipment	Main heat source	What clearance protects	Typical failure if ignored
Inverter	Conversion losses, fins	Air in below, out above; shade	Derating, fan wear, capacitor ageing
Battery	Charge/discharge losses	Separation, ventilation, egress	Faster ageing; fire spread risk
DB / MDB	Breakers, busbars, terminals	Spacing, vents, 0.6–0.9 m to ceiling	Hot terminals, nuisance trips, insulation damage
Cables	I^2R in conductor	Spacing, tray layers, shade	Insulation ageing, overheated joints
Joints, isolators	Contact resistance	Correct torque, crimp, rating	Hot spot → arc → fire

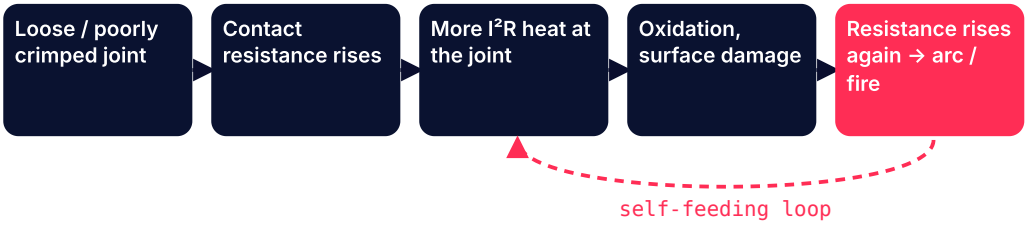
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EVIDENCE & CASES

What real incidents show

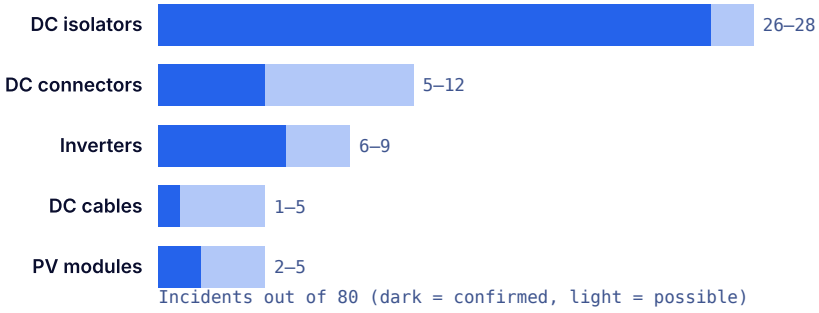
PV systems are safe overall: TÜV Rheinland and Fraunhofer ISE found that fewer than 0.006% of PV plants in Germany caused a fire [18]. The value of the studies is in *where* the few fires start — almost always at a joint, switch or connector where heat builds up, and mostly through installation mistakes.

The link between heat and fire



How a bad connection becomes a fire: heat raises resistance, which raises heat — the Schneider Electric white paper describes this thermal runaway ending in complete connection failure [22].

Case 1 · United Kingdom — BRE National Solar Centre (80 incidents)



DC isolators were the largest single risk, about 30% of incidents [17]. Some isolators were models designed for AC, or wrongly specified [17].

Case 2 · Germany — TÜV Rheinland study (210 PV fires)

Root causes were grouped as installation mistakes (badly mated or poorly crimped DC connectors, missing strain relief), product failures, external influences (animals, lightning) and planning failures; the "human factor" was the main source [18]. A badly crimped connector has a thinner contact area that raises its thermal resistance [18]. The related Fraunhofer/TÜV paper found aged DC switches with increased resistance and thermal damage [20].

Case 3 · Western Australia — DC isolators

As many as 50% of PV-caused fires in WA between 2009 and 2015 started at rooftop DC isolators [19].

Case 4 · Sweden — 2026 study of national fire statistics

DC cables and connectors were the most common ignition source, about a quarter of incidents; about two-thirds of incidents damaged only PV equipment [21]. DC switches and AC-side electrical units were other frequent ignition sources [21].

Case 5 · Brazil — inverters running too hot (180 kW plant)

Monitoring at a university plant showed inverters operating on average 30°C above the manufacturer's recommended limit, causing thermal derating and energy losses; the study tested forced ventilation as a fix [23].

Case 6 · All electrical installations — loose connections

One major insurer estimates about 25% of all major electrical failures are due to loose or faulty connections [22]; NFPA 70B suggests a thermographic survey of commercial and industrial buildings at least once a year [25].

HOW · SOLAR ESS PRACTICE

Getting it right: design, install, maintain

Stage	Air clearance	Working space	Heat dissipation
Design	Choose parts rated for DC and the system voltage (isolators, fuses, SPD); outdoor boxes rated for damp, dusty conditions	Mark the 0.75 m × depth × 2.0 m zone on the drawing; doors must open 90°	Place inverters and batteries on a shaded, ventilated wall; check derating temperature in the manual; size cables with temperature and grouping factors
Install	Correct lugs and ferrules; no loose strands; keep cables off busbars; tool-crimped, fully mated DC connectors of the same make	No shelves or storage in the zone; board top 0.9 m from a combustible ceiling (0.6 m non-combustible)	Respect the manual's gaps (often 300–500 mm top/bottom); never stack inverters vertically without spacing; 30 cm off sun-heated metal; no sealed cabinets
Commission	Insulation resistance test; torque every terminal to the maker's value and mark it	Photograph the clear zone for the handover file	Thermal scan at high load; record inverter internal temperature and any derating events
Maintain	Check glands and seals; clean dust; look for tracking marks and discoloured plastic	Remove anything stored in front of boards	Clean fins and filters; annual thermography of DC isolators, connectors, MDB terminals; compare monitoring data in the hot season

Five quick checks on any site

1. Can every door open fully, with 0.75 m clear in front?
2. Is there free air above and below each inverter?
3. Is the inverter or battery in direct afternoon sun?
4. Are DC isolators DC-rated, sealed and not hot to the touch?
5. Does monitoring show power clipping on hot afternoons?

Red flags

- Inverters stacked directly above each other
- Board or battery inside a closed cupboard or under a stair
- Brown or melted plastic at a terminal or connector
- Mixed-brand MC4 connectors or hand-crimped plugs
- Many circuits bundled in one conduit on a hot roof

Bottom line

Air clearance keeps voltage where it belongs; working space keeps people safe; heat clearance keeps every part inside its rating. All three are cheap at the design stage and expensive to fix later. The incident data point the same way: most PV fires begin at joints and switches that ran hot because of how they were chosen or fitted [17,18,21].

Prepared by Solar ESS Working Group. This report summarises public standards and studies for general guidance; for any installation, the equipment manual, the current EIT standard and PEA/MEA requirements take precedence. All qualifications and certificates will be furnished upon request or upon completion of contract.

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